

Multi-Objective Reinforcement Learning for RNA Inverse Folding

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Code repository: <https://github.com/atakmt/RL-Project>

1 Introduction

Ribonucleic acid (RNA) is a fundamental biomolecule involved in a wide range of critical biological processes, including the transcription and translation of genetic information, cellular differentiation and development, enzymatic catalysis, and the regulation of gene expression. Beyond its natural roles, RNA has also found significant applications in synthetic and therapeutic contexts, with engineered RNA molecules being employed in areas such as mRNA-based vaccines, RNA aptamers, genome editing tools, biosensing technologies, ribozymes, and riboswitches (Pardi et al., 2018; Keefe et al., 2010; Doudna and Charpentier, 2014; Breaker, 2018; Wachsmuth et al., 2013). The functional versatility of RNA in these contexts stems directly from its capacity to fold into intricate, sequence-determined three-dimensional shapes that mediate molecular recognition, catalysis, and ligand binding. Indeed, beyond their well-established roles, non-coding RNAs (ncRNAs) have been implicated in the regulation of epigenetic marks, alternative splicing, and the scaffolding of large macromolecular complexes (Vandivier et al., 2016), and their dysregulation has been linked to neurodegenerative disorders including Parkinson’s and Alzheimer’s diseases (Gstir et al., 2014; Kaushik et al., 2018). The ability to design RNA molecules with prescribed functions is therefore central to a growing range of biotechnological and therapeutic endeavors (Delebecque et al., 2011; Guo et al., 2010; Meyer et al., 2015).

At its most basic level, an RNA molecule is a sequence over the four-letter alphabet $\{A, C, G, U\}$, referred to as its primary structure. Function, however, is determined not by the sequence itself but by the three-dimensional fold it adopts under the intrinsic thermodynamic properties of base pairing and stacking interactions (Tinoco and Bustamante, 1999). The set of canonical Watson–Crick (GC, CG, AU, UA) and wobble base pairs (GU, UG) formed in the folded molecule defines its secondary structure, an intermediate abstraction that captures most of the free-energy contributions of folding while remaining computationally tractable (Zuker and Stiegler, 1981). Because tertiary structure prediction and design are often built directly on top of accurate secondary structures (Kerpedjiev et al., 2015; Reinharz et al., 2012), efficient algorithms for designing sequences that fold into prescribed secondary structures remain a central problem in RNA engineering.

Given this, the central problem of RNA inverse folding is to invert the structure-determining map: given a desired secondary structure ω , design a primary sequence φ that will fold into it. Figure 1 illustrates the three objects involved — the target structure ω written in dot-bracket notation (a), a candidate primary sequence $\varphi \in \{A, C, G, U\}^{|\omega|}$ (b), and the secondary structure $\mathcal{F}(\varphi)$ produced by folding that sequence with a thermodynamic minimum-free-energy folder $\mathcal{F}$ (c). A sequence is a valid solution to the inverse folding problem when $\mathcal{F}(\varphi) = \omega$. The remainder

of this paper formalises this problem (Section 3), extends it to a multi-objective setting that additionally accounts for GC-content, homopolymer runs, and thermodynamic stability, and develops a reinforcement-learning approach to solve it.

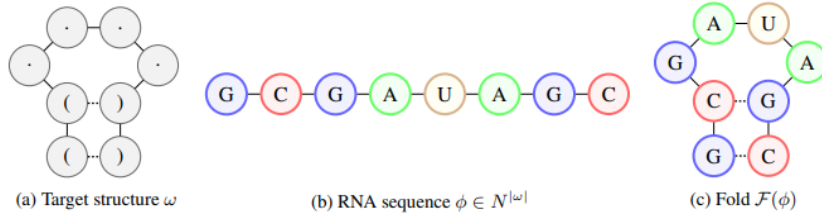


Figure 1: The RNA inverse folding problem. Given a target secondary structure ω (a), specified in dot-bracket notation, the goal is to design a primary nucleotide sequence $\varphi \in \{A, C, G, U\}^{|\omega|}$ (b) such that the folding oracle $\mathcal{F}$ returns $\mathcal{F}(\varphi) = \omega$ (c). Throughout this work, $\mathcal{F}$ is the Zuker minimum-free-energy folder as implemented in the ViennaRNA package (Lorenz et al., 2011).

2 Related Work

2.1 The RNA Inverse Folding Problem and Its Computational Hardness

RNA inverse folding has been studied for nearly three decades since the seminal RNAinverse algorithm of Hofacker et al. (1994), yet its formal complexity was only recently settled. Bonnet et al. (2020) showed that even in a simplified base-pair maximization energy model, the decision version of a mildly constrained inverse folding problem is NP-hard, and the problem remains NP-hard both in its general formulation and under simplified energy models. This intractability formally justifies the long-standing reliance on heuristics, metaheuristics, and learning-based methods, and motivates our framing of RNA design as a combinatorial optimization problem over a search space of size 4^n that cannot be exhaustively explored. Critically, the hardness result also implies that any method scaling to practical sequence lengths must accept approximate solutions, which directly motivates the use of stochastic, policy-based methods such as reinforcement learning that explore the search space efficiently without enumerating it.

2.2 Classical Heuristic and Metaheuristic Approaches

The earliest computational approaches to RNA inverse folding were grounded in local search and adaptive random walks. RNAinverse (Hofacker et al., 1994) initialized random sequences and iteratively modified mismatched positions, while INFO-RNA (Busch and Backofen, 2006) improved on this by initializing from the minimum-energy sequence for the target structure. However, this strategy was later shown to be problematic: starting local search from the minimum-energy sequence produces unrealistically high GC-contents in designed sequences, motivating the development of IncaRNA, which controls GC-content through an adaptive sampling strategy (Reinharz et al., 2013).

Subsequent global methods broadened the search via population-based metaheuristics. MODENA (Taneda, 2011) introduced multi-objective evolutionary optimization for RNA design, and its multi-target extension explicitly uses GC-content alongside MFE, ensemble defect, and structural similarity as objective functions, noting that biased GC content easily appears in designed sequences when not penalized. The antaRNA family applied ant-colony optimization and explicitly supported user-specified GC-content constraints; in its pseudoknot-aware extension, antaRNA returns sequences whose minimum free energy structure matches the target with a specified GC-content (Kleinkauf et al., 2015). MCTS-RNA (Yang et al., 2017) framed inverse folding as Monte Carlo tree search, and later, SAMFEO integrated ensemble-level information

for iterative sampling. Early search-based methods such as RNAinverse, MCTS-RNA, and antaRNA relied on heuristic optimization strategies, while later enhancements like IncaRNAion and SAMFEO integrated probabilistic sampling and ensemble-level insights.

These classical methods establish two important precedents for our work. First, GC-content has been recognized as a first-class design objective for over a decade; however, most modern deep learning approaches have regressed to single-objective structural optimization. Second, while MODENA and antaRNA jointly handle structure and GC-content, their objective-combination schemes are static across the optimization run, and they do not naturally accommodate synthesizability constraints such as homopolymer run penalties, nor curriculum-style annealing between competing objectives. Our approach shares with these classical methods the per-target design paradigm — a dedicated agent is trained from scratch for each target structure — but introduces a four-component composite reward and an adaptive weight curriculum on top of that paradigm.

2.3 Deep Reinforcement Learning for RNA Design

The current wave of deep learning approaches began with two parallel works in 2018–2019. Eastman et al. (2018) used reinforcement learning to perform local search, treating an existing sequence as state and learning a policy to flip individual nucleotides; in the LEARN framework this approach is referred to as RL-LS. In contrast, LEARN (Runge et al., 2019) introduced a generative RL formulation: starting from an empty sequence, the agent sequentially places nucleotides at each position, conditioned on a window of the target dot-bracket structure, and receives a Hamming-distance-based terminal reward after folding the completed sequence with ViennaRNA. LEARN further introduced Meta-LEARN, which trains a single policy across 65,000 RNA design tasks via asynchronous PPO updates, yielding a pre-trained agent applicable to new targets out of the box. Joint architecture and hyperparameter search via BOHB enabled LEARN to achieve state-of-the-art results on the Eterna100 and Rfam-Taneda benchmarks, while Meta-LEARN reached the previous state-of-the-art performance up to $1765\times$ faster.

Several follow-up works have extended this generative RL framework. libLEARN (Runge et al., 2024) reformulates RNA design as a constraint satisfaction problem allowing arbitrary sequence and structure motif constraints, and notably begins to incorporate auxiliary objectives. When designing riboswitch candidates with desired GC-contents, libLEARN provides up to 47% more candidates that satisfy the design criteria, and can design at the margins of possible GC-contents such as 0.3 and 0.6. This is the closest precedent for our work, but libLEARN still does not handle homopolymer runs or explicitly maximize MFE magnitude as a thermodynamic stability objective, nor does it employ curriculum-based weight annealing between objectives. DRAG takes a different direction: it builds hierarchical design environments using graph neural networks, with particular advantage on long and intricate structures with significant depth (Li et al., 2025). Most recently, Struct2SeQ (He and Sun, 2026) returns to the value-based paradigm: it uses deep Q-learning with dense, chemically informed rewards derived from neural predictors (RibonanzaNet) for both folding and SHAPE chemical reactivity, and formulates RNA design as sequential decision-making validated on OpenKnot 240mer pseudoknot challenges. Struct2SeQ validates two of our design choices: that DQN augmented with dense intermediate rewards is competitive with policy-gradient methods on this problem, and that position-wise reward shaping can effectively address the long-horizon credit assignment problem inherent to nucleotide-by-nucleotide generation.

2.4 Multi-Objective and Biophysically-Aware RNA Design

A separate line of recent work argues that purely structural objectives produce sequences that, while geometrically correct, are biophysically deficient. RiboPO introduces reinforcement learning from physical feedback for RNA tertiary inverse folding, explicitly incorporating thermodynamic stability alongside geometry and reframing inverse folding as a multi-objective preference

optimization problem (Sun et al., 2025). The motivation mirrors our own: single-objective optimization cannot fully address the multifaceted challenges inherent to RNA structural stability, where geometric fidelity must be balanced against thermodynamic preferences and conformational robustness. RiboPO fine-tunes a pre-trained generative model (gRNAd) via Direct Preference Optimization on preference pairs constructed from composite physical criteria. RIDER (Hu et al., 2026) similarly fine-tunes a GNN-based diffusion model with policy gradient using four task-specific reward functions based on 3D self-consistency metrics. Both works, however, operate on tertiary (3D) structure rather than secondary structure, use offline preference optimization or fine-tuning of pre-trained generative models rather than online RL from scratch, and target structural-quality metrics such as TM-score and pLDDT rather than synthesizability constraints. Our work occupies a different niche: online RL on the well-established Eterna100 secondary-structure benchmark, combining four heterogeneous objectives (structural Hamming distance, GC-content fitness, homopolymer avoidance, and normalized MFE) under an adaptive curriculum weighting schedule.

The mRNA design literature provides additional motivation for joint optimization of structural and biophysical properties. LinearDesign (Zhang et al., 2023) frames mRNA design as a lattice parsing problem: a principled mRNA design algorithm must optimize structural stability and codon usage to enhance protein expression, but the design problem is extremely challenging owing to the exponentially large search space, with each amino acid encoded by triplet codons. Although LinearDesign is restricted to protein-coding regions and its codon-usage objective (CAI) does not apply to the broader Eterna100 benchmark, it establishes that real-world RNA design is fundamentally multi-objective, and that structural matching alone is insufficient for therapeutic applicability. Leppek et al. (2022) reinforce this from the experimental side, demonstrating that mRNA stability and translation efficiency depend jointly on structure, GC composition, and the avoidance of problematic sequence motifs.

2.5 Why Reinforcement Learning Is the Right Tool

Three properties of the RNA inverse folding problem make reinforcement learning particularly suitable, and these properties are amplified rather than diminished in our multi-objective formulation. First, the reward signals are fundamentally non-differentiable: ViennaRNA’s Zuker folding algorithm is a dynamic-programming routine implemented in C, GC-content is a piecewise-linear function of discrete nucleotide counts, and homopolymer run length is a max over contiguous subsequences. None of these admit useful gradients with respect to the discrete nucleotide variables. As noted in recent work combining diffusion with RL, existing approaches including diffusion-based methods cannot effectively handle non-differentiable structural objectives, while RL excels by using policy-driven reward optimization to navigate complex, non-gradient-based objectives. This rules out end-to-end differentiable training and favors policy-gradient (PPO) or value-based (DQN) methods.

Second, the problem has natural sequential structure: nucleotides are placed one at a time, the partial sequence and target structure form a well-defined state, and rewards can be specified at both intermediate steps (local base-pair correctness, as in Struct2SeQ and our DQN reward shaping) and terminal steps (the global Hamming distance after folding, as in LEARN). This MDP framing has been validated repeatedly across the LEARN family, Struct2SeQ, DRAG, and the local-search formulation of Eastman et al. (2018), and provides a clean scaffold for both value-based and policy-gradient algorithms.

Third, and most importantly for our contribution, RL provides a unified scalar interface for combining heterogeneous objectives. The classical multi-objective tools (MODENA, antaRNA) handle two or three objectives via genetic operators or pheromone updates, but their objective-combination schemes are static throughout the optimization run. Reinforcement learning supports curriculum-style annealing of objective weights, allowing the agent to begin from a structurally-dominant regime and progressively incorporate biophysical penalties as the policy

stabilizes.

2.6 Positioning of Our Contribution

In light of the above, our work fills a specific and well-defined gap. LEARNNA and Meta-LEARNNA optimize only structural fidelity. libLEARNNA adds GC-content but does not handle homopolymer runs or MFE magnitude, and uses static rather than adaptive weighting. RiboPO and RIDER address multi-objective design but operate on tertiary structure with offline preference optimization or generative-model fine-tuning, not online RL on secondary-structure benchmarks. Struct2SeQ uses DQN with dense intermediate rewards but focuses on SHAPE chemical reactivity rather than synthesizability constraints. To the best of our knowledge, no prior work performs (a) online RL on secondary-structure inverse folding with (b) a four-component reward combining structural Hamming distance, GC-content fitness, homopolymer avoidance, and normalized MFE, (c) under an adaptive curriculum weighting schedule, and (d) with a controlled head-to-head comparison between value-based (DQN with reward shaping) and policy-gradient (PPO) paradigms. By targeting the standard Eterna100 benchmark and building on the open-source LEARNNA Gymnasium environment, our work also remains directly comparable to the established secondary-structure RL literature, allowing the additional sequence-quality objectives to be evaluated as proper extensions rather than orthogonal benchmarks.

3 Problem Definition

An RNA folding algorithm $\mathcal{F}$ maps a primary nucleotide sequence to a representation of its secondary structure. The RNA design problem, in its classical formulation (Hofacker et al., 1994), seeks an inverse of this mapping: given $\mathcal{F}$ and a target structure ω , find a sequence $\varphi \in \{A, C, G, U\}^{|\omega|}$ such that $\mathcal{F}(\varphi) = \omega$. Following LEARNNA (Runge et al., 2019) and the broader RNA design literature, we adopt the Zuker dynamic-programming algorithm (Zuker and Stiegler, 1981) as implemented in the ViennaRNA package (Lorenz et al., 2011) as our folding oracle $\mathcal{F}$, and represent target secondary structures using the standard dot-bracket notation, in which unpaired positions are denoted by dots and base-paired positions are marked by matching parentheses.

The classical formulation reduces RNA design to the minimization of a single structural loss function $\mathcal{L}_\omega(\mathcal{F}(\varphi))$ — typically the normalized Hamming distance between the predicted and target structures — over the discrete sequence space:

$$\varphi^* \in \arg \min_{\varphi \in \{A, C, G, U\}^{|\omega|}} \mathcal{L}_\omega(\mathcal{F}(\varphi)). \quad (1)$$

While this formulation captures the geometric fidelity requirement of inverse folding, it ignores several sequence-level properties that determine whether a designed molecule is biophysically stable, synthesizable, and suitable for downstream experimental use. We therefore extend the classical definition to a multi-objective setting.

Definition 1 (Multi-Objective RNA Design). *Given a folding algorithm $\mathcal{F}$ and a target secondary structure ω of length n , the multi-objective RNA design problem is to find a sequence $\varphi \in \{A, C, G, U\}^n$ that jointly minimizes a vector of design losses*

$$\mathbf{L}_\omega(\varphi) = (\mathcal{L}_\omega^{\text{struct}}(\varphi), \mathcal{L}^{\text{GC}}(\varphi), \mathcal{L}^{\text{homo}}(\varphi), \mathcal{L}^{\text{MFE}}(\varphi)), \quad (2)$$

where the four components are defined as follows.

The **structural loss** $\mathcal{L}_\omega^{\text{struct}}(\varphi) = d_H(\mathcal{F}(\varphi), \omega)/n$ is the normalized Hamming distance between the dot-bracket encoding of the predicted minimum-free-energy fold $\mathcal{F}(\varphi)$ and the target ω , identical to the loss used in LEARNNA. A value of zero indicates perfect topological agreement.

The **GC-content loss** $\mathcal{L}^{\text{GC}}(\varphi)$ penalizes deviations of the sequence composition from a biologically plausible band. Letting $g(\varphi)$ denote the fraction of G and C nucleotides in φ , we define

$$\mathcal{L}^{\text{GC}}(\varphi) = \frac{\max(0, |g(\varphi) - 0.5| - 0.1)}{0.4}, \quad (3)$$

which is zero for sequences with GC content in the range $[0.4, 0.6]$ and increases linearly outside this range. Extreme GC composition has been associated with RNA misfolding, aggregation, and reduced solubility in therapeutic contexts (Leppek et al., 2022; Reinharz et al., 2013).

The **homopolymer loss** $\mathcal{L}^{\text{homo}}(\varphi)$ penalizes contiguous runs of identical nucleotides exceeding a synthesizability threshold k (we use $k = 4$). Let $\mathcal{R}(\varphi)$ denote the multiset of maximal same-nucleotide runs in φ ; we define

$$\mathcal{L}^{\text{homo}}(\varphi) = \frac{1}{n} \sum_{r \in \mathcal{R}(\varphi)} \max(0, |r| - k). \quad (4)$$

Long homopolymer stretches impair chemical synthesis fidelity, cause sequencing errors, and reduce structural diversity (Leppek et al., 2022).

The **thermodynamic stability loss** $\mathcal{L}^{\text{MFE}}(\varphi)$ rewards deep free-energy minima. Let $E(\varphi)$ denote the minimum free energy (in kcal/mol) of the predicted fold returned by $\mathcal{F}$; we define

$$\mathcal{L}^{\text{MFE}}(\varphi) = -\frac{|E(\varphi)|}{n}. \quad (5)$$

Sequences with more negative MFE per nucleotide fold into their target topology with higher thermodynamic confidence and are less prone to misfolding into competing conformations (Sun et al., 2025).

A sequence φ is *Pareto-optimal* if no other sequence φ' simultaneously achieves a lower value in every component of $\mathbf{L}_\omega$ while being strictly lower in at least one. The full Pareto frontier is generally intractable to enumerate, given the NP-hardness of even the single-objective problem (Bonnet et al., 2020). We therefore adopt the standard scalarization approach (Roijers et al., 2013) and convert the vector-valued problem to a sequence of scalar problems via weighted aggregation:

$$\varphi^*(\mathbf{w}) \in \arg \min_{\varphi \in \{A,C,G,U\}^n} \mathbf{w}^\top \mathbf{L}_\omega(\varphi), \quad \mathbf{w} = (\alpha, \beta, \gamma, \delta), \quad (6)$$

where the weight vector $\mathbf{w}$ controls the trade-off between objectives. To prevent the well-documented pathology of *objective dominance* — in which scalarized agents collapse onto whichever component is easiest to optimize early in training — we treat $\mathbf{w}$ not as a fixed hyperparameter but as a time-varying curriculum, detailed in Section 4.

Several remarks are in order. First, as in the single-objective formulation, the multi-objective problem does not generally admit a unique solution: many distinct sequences may achieve equivalent objective vectors, and we make no attempt to disambiguate among them. Second, our formulation strictly generalizes the LEARNNA design problem: setting $\mathbf{w} = (1, 0, 0, 0)$ recovers the structural-only objective exactly, allowing direct comparison with the single-objective baselines. Third, all four objectives are universally applicable to any RNA secondary structure in the Eterna100 benchmark (Anderson-Lee et al., 2016), regardless of whether the target represents a coding or non-coding region; this is in contrast to mRNA-specific design objectives such as the Codon Adaptation Index, which require the target to be a translatable open reading frame.

4 Methodology

4.1 Markov Decision Process Formulation

Building on the multi-objective design problem defined in Section 3, we cast RNA inverse folding as a finite-horizon episodic Markov Decision Process (MDP). The agent constructs a candidate

sequence nucleotide-by-nucleotide, with each episode terminating after exactly $n = |\omega|$ steps. Table 1 summarizes the MDP components.

Element	Definition
State s_t	Partial sequence to step t , target structure encoding, local pairing context
Action a_t	Discrete nucleotide selection from $\{A, C, G, U\}$, $ \mathcal{A} = 4$
Transition	Deterministic; selected nucleotide is appended to the partial sequence
Reward	Potential-based shaping for $t < n$; compound reward at $t = n$
Horizon	Fixed at n steps
Discount γ	0.99

Table 1: MDP components for the multi-objective RNA design environment.

4.2 Observation Space

The state s_t is encoded as a continuous vector of dimension $7n + 10$, decomposed into a global context and a partner-aware local context.

Global context ($7n + 1$ dimensions). Placed nucleotides are one-hot encoded over 4 dimensions per position ($4n$ total), with unplaced positions represented as zero vectors. The target dot-bracket structure is one-hot encoded over 3 dimensions per position ($3n$ total), and a scalar progress indicator $t/n \in [0, 1]$ informs the agent of its location in the rollout. We use one-hot rather than ordinal encoding because the latter introduces spurious magnitude relationships that prevented convergence in preliminary experiments.

Partner-aware local context (9 dimensions). At each step, the agent additionally observes a 9-dimensional vector encoding the current target character (one-hot, 3 dims), whether the position is paired (1 dim), whether the base-pair partner has already been placed (1 dim), and the partner’s nucleotide identity when available (one-hot, 4 dims). This local context directly exposes complementarity information at closing-bracket positions, removing the need for the agent to implicitly recover partner identity from the full sequence vector — a task whose difficulty grows with sequence length. In ablations, this addition raised DQN’s structural reward from 0.27 to 0.46 on Puzzle 54 (92 nt) and from 0.55 to 0.82 on Puzzle 65 (40 nt).

4.3 Reward Function

Terminal reward. At step $t = n$, the completed sequence φ is folded via ViennaRNA’s Zuker algorithm and scored by the compound reward

$$R(\varphi) = \alpha R_{\text{struct}}(\varphi) + \beta R_{\text{GC}}(\varphi) - \gamma P_{\text{homo}}(\varphi) + \delta R_{\text{MFE}}(\varphi), \quad (7)$$

where the reward-form components are obtained from the losses in Section 3 via $R_{\text{struct}} = 1 - \mathcal{L}_{\omega}^{\text{struct}}$, $R_{\text{GC}} = 1 - \mathcal{L}^{\text{GC}}$, $P_{\text{homo}} = \mathcal{L}^{\text{homo}}$, and $R_{\text{MFE}} = -\mathcal{L}^{\text{MFE}}$. The weight vector $\mathbf{w} = (\alpha, \beta, \gamma, \delta)$ is annealed according to the curriculum described below.

Intermediate shaping. Since structural feedback is only available after folding the completed sequence, inverse folding is a sparse-reward problem that suffers from severe long-horizon credit assignment for $n > 30$. We employ potential-based reward shaping (Ng et al., 1999), which preserves the optimal policy while injecting dense intermediate signal. Defining the potential function over paired positions placed so far,

$$\Phi(s_t) = \frac{|\{(i, j) \in \mathcal{P} : i, j < t \text{ and } \text{WC}(s_i, s_j)\}|}{|\{(i, j) \in \mathcal{P} : i, j < t\}|}, \quad (8)$$

the shaping reward at each intermediate step is $F(s, a, s') = \kappa(\gamma\Phi(s') - \Phi(s))$, with $\kappa = 0.1$ chosen so that shaping does not dominate the terminal reward. Φ is set to zero at the absorbing state to preserve theoretical equivalence with the unshaped MDP.

4.4 Three-Phase Curriculum

Static scalarization is known to produce *objective dominance* in multi-objective RL, where the agent collapses onto whichever component yields the strongest early gradient (Roijers et al., 2013). We therefore train under a three-phase weight schedule. In **Phase A** (0–30% of training), we set $\alpha = 1.0$, $\beta = 0.7\beta^*$, $\gamma = \gamma^*$, $\delta = 0$: the structural signal is at full strength, the GC and homopolymer penalties are partially active to prevent compositional collapse, and MFE is deferred. In **Phase B** (30–70%), all weights are linearly interpolated to their target values $(\alpha^*, \beta^*, \gamma^*, \delta^*)$. In **Phase C** (70–100%), weights are held fixed at their targets.

The GC-content collapse pathology. A naive Phase A with $\beta = \gamma = \delta = 0$ — pure structural warmup — causes irreversible all-GC collapse. Because G–C pairs contribute three hydrogen bonds versus two for A–U, the agent rapidly discovers that all-GC sequences satisfy structural constraints most easily; policy entropy collapses around this deterministic selection. By the time β ramps up in Phase B, the GC reward landscape is flat at $R_{GC} = 0$ on the all-GC plateau, providing no gradient to pull the policy back. Maintaining $\beta = 0.7\beta^*$ and $\gamma = \gamma^*$ from step zero, combined with PPO’s entropy bonus (`ent_coef` = 0.02), prevents this attractor from forming.

4.5 Algorithms

We compare two fundamentally different RL paradigms on the same environment and reward.

PPO (on-policy, policy gradient) (Schulman et al., 2017). A [128, 128] MLP policy with clipped surrogate objective, $n_{\text{steps}} = 128$, batch size 64, 10 epochs per rollout, constant learning rate 3×10^{-4} , entropy coefficient 0.02, and $\gamma = 0.99$. Generalized Advantage Estimation (GAE) propagates credit across the entire n -step rollout in a single backward pass.

DQN (off-policy, value-based) (Mnih et al., 2015). A [256, 256] MLP Q-network with replay buffer size 50,000, batch size 128, 1,000-step random warmup, exploration fraction 0.6 with final $\varepsilon = 0.08$, soft target updates ($\tau = 0.005$, update interval 500), and $\gamma = 0.99$. The intermediate shaping reward is essential here, since 1-step TD bootstrapping requires $\mathcal{O}(n)$ updates to propagate terminal reward to the first step.

Credit assignment. The two algorithms differ fundamentally in how reward signal reaches early steps of a rollout. PPO’s GAE assigns advantage to every step in one backward pass, while DQN must propagate Q-values one step at a time through replay. For $n > 30$ this gives PPO a substantial structural advantage that intermediate shaping is designed to partially offset for DQN.

4.6 Training Budget

Episode counts scale with target length so that longer, harder structures receive proportionally more compute:

$$E(n) = E_0 \cdot \max\left(1, 1 + \frac{n-30}{30}\right), \quad (9)$$

with $E_0 = 3,000$ for PPO and $E_0 = 5,000$ for DQN. Total environment interactions per target are $n \cdot E(n)$.

4.7 Experimental Protocol

Dataset. We select 15 target structures from the Eternal100-V2 benchmark (Anderson-Lee et al., 2016), spanning sequence lengths from 12 to 92 nucleotides and including hairpins, bulges, internal loops, multi-loops, branching structures, and a high-difficulty 7-way multiloop (Puzzle 54, 92 nt). Each target is solved independently: a dedicated PPO (or DQN) agent is trained from scratch for that single structure and saved as a target-specific specialist policy. The evaluation is therefore best understood as a per-puzzle benchmark of specialist policies rather than a transfer-learning study.

Grid search. We initially planned to evaluate three weight configurations $(\alpha^*, \beta^*, \gamma^*, \delta^*)$: a *balanced* setting (0.5, 0.2, 0.1, 0.2), a *structure-heavy* setting (0.6, 0.15, 0.1, 0.15), and a *thermodynamic-focused* setting (0.4, 0.2, 0.15, 0.25). Preliminary runs revealed that the three configurations produced near-identical training trajectories on every target, an artifact of the fixed seed combined with the structure-dominated Phase A schedule, which causes the per-target policy to converge before the target weights take effect. We therefore report results only for the balanced configuration in this progress report and defer a more strongly differentiated grid search to future work (Section 6).

Evaluation. Training metrics are logged per objective as rolling means over the last 50 episodes. Final evaluation uses deterministic policies (PPO with `deterministic=True`, DQN with $\varepsilon = 0$). A target counts as *solved* when the MFE-folded structure has Hamming distance zero from the target ($R_{\text{struct}} = 1$); for solved sequences, we additionally report R_{GC} , P_{homo} , and R_{MFE} to characterize biophysical quality.

5 Results

5.1 Setup

We report results for both PPO and DQN under the balanced weight configuration $(\alpha^*, \beta^*, \gamma^*, \delta^*) = (0.5, 0.2, 0.1, 0.2)$ with seed 42. Each algorithm is trained independently on every one of the 15 target structures using the per-target specialist setup described in Section 4. Training budgets follow Equation (9): episodes per target scale linearly with target length above $n = 30$, with $E_0 = 3,000$ for PPO and $E_0 = 5,000$ for DQN (the larger DQN budget compensates for its 1-step TD credit-assignment overhead). All metrics are reported at the per-puzzle level: R_{struct} (mean over the last 50 episodes), best R_{struct} achieved during training, and the success rate (fraction of last-50 episodes for which $R_{\text{struct}} = 1$).

5.2 Per-Puzzle Training Results

Table 2 reports per-puzzle final R_{struct} , best R_{struct} , and success rate for both algorithms across the 15 targets.

5.3 PPO vs. DQN: Aggregate Comparison

Table 3 summarizes the head-to-head comparison between the two algorithms. PPO dominates DQN on all four primary metrics, despite training for $\sim 60\%$ as many base episodes per target and finishing the full 15-target sweep in roughly half the elapsed time.

Three findings stand out. **First**, on the two shortest puzzles (P1, P8) PPO converges to and maintains a 100% success rate, while DQN oscillates around the optimum at 38% and 56%, respectively. This is a behavioural signature of DQN’s $\varepsilon = 0.08$ residual exploration floor: even after the Q-network has identified the optimal action at each position, the ε -greedy policy

Puzzle	Name	Len	PPO			DQN		
			R_{struct}	Best	Success	R_{struct}	Best	Success
P1	Simple Hairpin	18	1.000	1.000	100%	0.844	1.000	38%
P8	G-C Placement	12	1.000	1.000	100%	0.830	1.000	56%
P10	Frog Foot	45	0.648	0.822	0%	0.508	1.000	0%
P13	Square	67	0.876	1.000	0%	0.553	1.000	0%
P15	Small and Easy 6	30	0.845	0.867	0%	0.486	0.800	0%
P23	Shortie 4	17	0.646	0.765	0%	0.401	0.765	0%
P25	The Ministry	62	0.567	0.903	0%	0.496	0.774	0%
P26	Stickshift	26	0.792	1.000	0%	0.721	1.000	2%
P30	Corner Bulge Training	31	0.888	1.000	62%	0.734	1.000	10%
P40	Tripod5	38	0.785	0.947	0%	0.537	0.947	0%
P41	Shortie 6	35	0.542	0.771	0%	0.438	0.714	0%
P45	5-Adj. Stack ML	63	0.679	1.000	0%	0.439	0.873	0%
P47	Misfolded Aptamer	50	0.823	1.000	0%	0.482	0.800	0%
P54	7-Multiloop	92	0.518	1.000	0%	0.461	0.783	0%
P65	Branching Loop	40	0.932	0.950	0%	0.595	0.950	0%

Table 2: Per-puzzle training results for PPO and DQN under the balanced weight configuration with seed 42. R_{struct} is the mean structural reward over the last 50 episodes; *Best* is the maximum attained at any point during training; *Success* is the fraction of last-50 episodes that recovered the target structure exactly. DQN was given a larger episode budget ($E_0 = 5,000$ versus 3,000 for PPO) yet underperforms PPO on every aggregate metric.

Metric	PPO	DQN
Mean final R_{struct} (across 15 puzzles)	0.769	0.568
Mean best R_{struct} (during training)	0.935	0.894
Mean success rate	10.8%	7.1%
Puzzles solved at $\geq 50\%$ success	3/15	1/15
Puzzles where $R_{\text{struct}} = 1$ was ever reached	8/15	6/15
Mean final R_{struct} on long targets ($n \geq 40$, 7 puzzles)	0.720	0.505
Base episode count E_0	3,000	5,000
Total elapsed time (15-puzzle sweep, seed 42)	~ 112 min	~ 213 min

Table 3: PPO versus DQN aggregate metrics under the balanced weight configuration. PPO outperforms DQN on final structural reward, success rate, and solved-puzzle count, despite a smaller per-target episode budget and roughly half the elapsed time.

continues to inject random nucleotide flips, repeatedly destroying a structurally-correct sequence in a way that PPO’s entropy-regularized policy does not.

Second, on the medium-length puzzles where PPO briefly reaches $R_{\text{struct}} = 1$ before drifting (P13, P26, P45, P47, P54), DQN similarly touches the optimum on a subset of these targets but converges to a much lower final R_{struct} . The gap between best and final reward is wider for DQN ($0.894 - 0.568 = 0.326$) than for PPO ($0.935 - 0.769 = 0.166$), indicating that DQN’s Q-value bootstrapping is more easily destabilized by the Phase-C activation of the MFE and GC objectives.

Third, the empirical $n > 30$ advantage of PPO predicted in Section 4 (Algorithms: Credit assignment) is confirmed: averaged over the seven puzzles with $n \geq 40$ (P10, P13, P25, P45, P47, P54, P65), PPO achieves $R_{\text{struct}} = 0.720$ versus DQN’s 0.505, a gap of 0.215. PPO’s GAE assigns advantage across the full n -step rollout in one backward pass, whereas DQN’s 1-step TD updates require $\mathcal{O}(n)$ replay cycles to propagate the terminal Hamming-distance reward back to the first nucleotide placement — a bottleneck that intermediate reward shaping alleviates but

does not eliminate.

5.4 Discussion

Reliable convergence on short structures (PPO). For both the 18-nt and 12-nt targets (P1, P8), PPO reaches $R_{\text{struct}} = 1$ within the first 15–20% of training and maintains a 100% success rate through Phases B and C, indicating that the curriculum’s Phase-C reintroduction of biophysical objectives does not destabilize an already-solved structure when the target is short enough that the policy enters Phase B with high confidence. DQN reaches the same optima (Best $R_{\text{struct}} = 1.000$ for both P1 and P8) but cannot lock onto them due to ε -greedy noise, as discussed in Section 5.3.

Optimum-touching without lock-in on medium targets. For both algorithms, several puzzles (P13, P26, P30, P45, P47, P54 for PPO; additionally P10 for DQN) reach $R_{\text{struct}} = 1$ at least once during training but fail to maintain it in the last 50 episodes. The training logs show the typical pattern: R_{struct} peaks during Phase B, then the Phase-C activation of δ (MFE) and the full β (GC) causes the policy to drift away from the structurally-perfect solution toward sequences that improve composite reward but lose structural fidelity. Only P30 (Corner Bulge Training, 31 nt) escapes this pattern — PPO holds a 62% success rate, DQN only 10%. This is consistent with the objective-dominance pathology that the curriculum (Section 4.4) was designed to mitigate, suggesting that the current Phase-C weights pull too strongly relative to the structural component for medium-length targets, and that DQN is substantially more vulnerable to this drift than PPO.

Plateau on long and topologically dense targets. For P10 (Frog Foot, three hairpins on 45 nt), P25 (The Ministry, 62 nt) and P41 (Shortie 6, 35 nt with four short hairpins), PPO never reaches $R_{\text{struct}} = 1$ in its final 50 episodes, and DQN performs uniformly worse on these targets. They are precisely the structures that combine moderate length with repeated motifs, requiring the policy to coordinate identical sub-decisions across non-contiguous regions of the sequence — a credit-assignment problem that the partner-aware observation (Section 4.2) only partially addresses, and to which DQN’s 1-step bootstrapping is especially vulnerable.

Aggregate finding. Across the 15 specialist policies, PPO outperforms DQN on every aggregate metric: mean final R_{struct} (0.769 vs 0.568), mean best R_{struct} (0.935 vs 0.894), success rate (10.8% vs 7.1%), and solved-puzzle count (3 vs 1). The gap between mean best and mean final R_{struct} (0.166 for PPO, 0.326 for DQN) remains the main quantitative finding for both algorithms: the policy is capable of reaching the structural optimum for many targets, but cannot stably hold it under the current Phase-C joint-optimization regime — and DQN suffers this collapse roughly twice as severely as PPO. Combined with the elapsed-time advantage (PPO completes the 15-puzzle sweep in $\sim 53\%$ of DQN’s runtime) and the credit-assignment evidence from the $n \geq 40$ subset, these results identify PPO as the clearly preferable algorithm for this problem under the current setup, and motivate the curriculum refinements proposed in Section 6 as the primary lever for closing the remaining best-versus-final gap.

6 Future Work

The results in Section 5 point to four concrete directions that will form the focus of the remaining project period, each aimed at closing the gap between the mean best R_{struct} (0.935) and the mean converged R_{struct} (0.769) observed across the 15 targets.

Full multi-objective evaluation of solved puzzles. The comparisons reported in Section 5 are deliberately restricted to the structural reward R_{struct} , because this is the only objective for which we currently have a per-puzzle, per-algorithm signal that is both well-calibrated and directly interpretable. This restriction, however, leaves the central question of the multi-objective formulation unanswered: do the sequences that achieve $R_{\text{struct}} = 1$ also satisfy the GC-content, homopolymer, and thermodynamic stability criteria simultaneously? For the final report we will extend `evaluate_deterministic.py` to (i) generate a canonical sequence per puzzle by running each saved specialist policy with stochasticity disabled, (ii) compute R_{struct} , $\mathcal{L}^{\text{GC}}$, $\mathcal{L}^{\text{homo}}$, and R_{MFE} on that exact sequence rather than as rolling training-time averages, and (iii) declare a puzzle *fully solved* only when all four criteria are simultaneously satisfied — specifically $R_{\text{struct}} = 1$, $g(\varphi) \in [0.4, 0.6]$, no homopolymer run exceeding $k = 4$ nucleotides, and $|E(\varphi)|/n$ above a domain-appropriate threshold τ_{MFE} . The per-puzzle and aggregate tables of the final report will then present both PPO and DQN side-by-side across all four objectives, will list the canonical sequences produced for the fully-solved targets, and will report both the structural and the fully-solved counts as headline metrics — restoring the multi-objective framing that motivates the entire project.

Curriculum re-tuning to preserve structural fidelity in Phase C. The dominant failure mode is policy drift away from structurally-perfect solutions once the MFE component is activated. We will (i) shorten the Phase-A fraction so that the target weights are reached earlier, while the policy still has high exploration entropy; (ii) introduce a soft floor on α in Phase C so that the structural gradient cannot be fully overpowered by the biophysical terms; and (iii) re-instate the grid search with more strongly differentiated weight settings (e.g. a structure-only control at $\mathbf{w} = (1.0, 0, 0, 0)$ and biophysics-heavy configurations such as $(0.4, 0.5, 0.1, 0)$ and $(0.4, 0, 0.1, 0.5)$) and additional seeds, so that the trade-off frontier becomes empirically observable rather than collapsing into a single trajectory.

Hyperparameter and budget tuning for long targets. Targets above $n = 50$ exhaust their training budget before R_{struct} stabilizes. We will conduct a small-scale hyperparameter sweep over PPO’s entropy coefficient, n_{steps} , and clip range, and over DQN’s exploration schedule and target-update interval, focused on the three hardest puzzles (P25, P54, P45). Episode budgets will be re-derived from empirical learning-curve slopes rather than the linear length heuristic of Equation (9).

Code and runtime optimization. Each call to ViennaRNA’s Zuker folder dominates the per-step elapsed time. We will (i) cache fold results for partial sequences during the rollout, (ii) move to a `SubprocVecEnv` of vectorized environments so multiple targets can be trained in parallel rather than sequentially, and (iii) profile and remove redundant observation-encoding work currently performed inside `LearnaEnv`. Together, these changes are expected to roughly halve the elapsed time required for a full 15-target run, freeing compute for the curriculum and hyperparameter sweeps above.

References

- Anderson-Lee, J., Fisker, E., Kosaraju, V., Wu, M., Kong, J., Lee, J., Lee, M., Zada, M., Treuille, A., & Das, R. (2016). Principles for predicting RNA secondary structure design difficulty. *Journal of Molecular Biology*, 428(5), 748–757. <https://doi.org/10.1016/j.jmb.2015.11.013>
- Bonnet, É., Rzazewski, P., & Sikora, F. (2020). Designing RNA secondary structures is hard.

- Journal of Computational Biology*, 27(3), 302–316. <https://doi.org/10.1089/cmb.2019.0420>
- Breaker, R. R. (2018). Riboswitches and translation control. *Cold Spring Harbor Perspectives in Biology*, 10(11), a032797. <https://doi.org/10.1101/cshperspect.a032797>
- Busch, A., & Backofen, R. (2006). INFO-RNA—a fast approach to inverse RNA folding. *Bioinformatics*, 22(15), 1823–1831. <https://doi.org/10.1093/bioinformatics/btl1194>
- Delebecque, C. J., Lindner, A. B., Silver, P. A., & Aldaye, F. A. (2011). Organization of intracellular reactions with rationally designed RNA assemblies. *Science*, 333(6041), 470–474. <https://doi.org/10.1126/science.1206938>
- Doudna, J. A., & Charpentier, E. (2014). The new frontier of genome engineering with CRISPR-Cas9. *Science*, 346(6213), 1258096. <https://doi.org/10.1126/science.1258096>
- Eastman, P., Shi, J., Ramsundar, B., & Pande, V. S. (2018). Solving the RNA design problem with reinforcement learning. *PLoS Computational Biology*, 14(6), e1006176. <https://doi.org/10.1371/journal.pcbi.1006176>
- Gstir, R., Schafferer, S., Scheideler, M., Misslinger, M., Griehl, M., Daschil, N., Humpel, C., Obermair, G. J., Schmuckermair, C., Striessnig, J., Flucher, B. E., & Hüttenhofer, A. (2014). Generation of a neuro-specific microarray reveals novel differentially expressed noncoding RNAs in mouse models for neurodegenerative diseases. *RNA*, 20(12), 1929–1943. <https://doi.org/10.1261/rna.047225.114>
- Guo, P., Coban, O., Snead, N. M., Trebley, J., Hoeprich, S., Guo, S., & Shu, Y. (2010). Engineering RNA for targeted siRNA delivery and medical application. *Advanced Drug Delivery Reviews*, 62(6), 650–666. <https://doi.org/10.1016/j.addr.2010.03.008>
- He, S., & Sun, Q. (2026). *Struct2SeQ: RNA inverse folding with deep Q-learning* [Preprint]. bioRxiv. <https://doi.org/10.64898/2026.01.16.700031>
- Hofacker, I. L., Fontana, W., Stadler, P. F., Bonhoeffer, L. S., Tacker, M., & Schuster, P. (1994). Fast folding and comparison of RNA secondary structures. *Monatshefte für Chemie*, 125(2), 167–188. <https://doi.org/10.1007/BF00818163>
- Hu, T., Cui, Y., Luo, B., & Li, K. (2026). *RIDER: 3D RNA inverse design with reinforcement learning-guided diffusion* [Preprint]. arXiv. <https://arxiv.org/abs/2602.16548>
- Kaushik, K., Sivadas, A., Vellarikkal, S. K., Verma, A., Jayarajan, R., Pandey, S., Sethi, T., Maiti, S., Scaria, V., & Sivasubbu, S. (2018). RNA secondary structure profiling in zebrafish reveals unique regulatory features. *BMC Genomics*, 19(1), 147. <https://doi.org/10.1186/s12864-018-4538-8>
- Keefe, A. D., Pai, S., & Ellington, A. (2010). Aptamers as therapeutics. *Nature Reviews Drug Discovery*, 9(7), 537–550. <https://doi.org/10.1038/nrd3141>
- Kerpedjiev, P., Höner zu Siederdissen, C., & Hofacker, I. L. (2015). Predicting RNA 3D structure using a coarse-grain helix-centered model. *RNA*, 21(6), 1110–1121. <https://doi.org/10.1261/rna.047522.114>
- Kleinkauf, R., Houwaart, T., Backofen, R., & Mann, M. (2015). antaRNA—Multi-objective inverse folding of pseudoknot RNA using ant-colony optimization. *BMC Bioinformatics*, 16(1), 389. <https://doi.org/10.1186/s12859-015-0815-6>

- Leppke, K., Byeon, G. W., Kladwang, W., Wayment-Steele, H. K., Kerr, C. H., Xu, A. F., et al. (2022). Combinatorial optimization of mRNA structure, stability, and translation for RNA-based therapeutics. *Nature Communications*, 13(1), 1536. <https://doi.org/10.1038/s41467-022-28776-w>
- Li, Y., Pan, X., Shen, H., & Yang, Y. (2025). DRAG: design RNAs as hierarchical graphs with reinforcement learning. *Briefings in Bioinformatics*, 26(2). <https://doi.org/10.1093/bib/bbaf106>
- Lorenz, R., Bernhart, S. H., Höner zu Siederdissen, C., Tafer, H., Flamm, C., Stadler, P. F., & Hofacker, I. L. (2011). ViennaRNA Package 2.0. *Algorithms for Molecular Biology*, 6(1), 26. <https://doi.org/10.1186/1748-7188-6-26>
- Meyer, S., Chappell, J., Sankar, S., Chew, R., & Lucks, J. B. (2015). Improving fold activation of small transcription activating RNAs (STARs) with rational RNA engineering strategies. *Biotechnology and Bioengineering*, 113(1), 216–225. <https://doi.org/10.1002/bit.25693>
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529–533. <https://doi.org/10.1038/nature14236>
- Ng, A. Y., Harada, D., & Russell, S. J. (1999). Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the 16th International Conference on Machine Learning (ICML 1999)* (pp. 278–287). Morgan Kaufmann.
- Pardi, N., Hogan, M. J., Porter, F. W., & Weissman, D. (2018). mRNA vaccines—a new era in vaccinology. *Nature Reviews Drug Discovery*, 17(4), 261–279. <https://doi.org/10.1038/nrd.2017.243>
- Reinharz, V., Major, F., & Waldispühl, J. (2012). Towards 3D structure prediction of large RNA molecules: an integer programming framework to insert local 3D motifs in RNA secondary structure. *Bioinformatics*, 28(12), i207–i214. <https://doi.org/10.1093/bioinformatics/bts226>
- Reinharz, V., Ponty, Y., & Waldispühl, J. (2013). A weighted sampling algorithm for the design of RNA sequences with targeted secondary structure and nucleotide distribution. *Bioinformatics*, 29(13), i308–i315. <https://doi.org/10.1093/bioinformatics/btt217>
- Rojers, D. M., Vamplew, P., Whiteson, S., & Dazeley, R. (2013). A survey of multi-objective sequential decision-making. *Journal of Artificial Intelligence Research*, 48, 67–113. <https://doi.org/10.1613/jair.3987>
- Runge, F., Stoll, D., Falkner, S., & Hutter, F. (2019). Learning to design RNA. In *Proceedings of the 7th International Conference on Learning Representations (ICLR 2019)*. <https://arxiv.org/abs/1812.11951>
- Runge, F., Franke, J., Fertmann, D., Backofen, R., & Hutter, F. (2024). Partial RNA design. *Bioinformatics*, 40(Supplement_1), i437–i445. <https://doi.org/10.1093/bioinformatics/btae222>
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). *Proximal policy optimization algorithms* [Preprint]. arXiv. <https://arxiv.org/abs/1707.06347>
- Sun, M., Cao, H., Zhang, Z., Wei, C., Wang, L., Jia, T., Liu, Z., Fu, T., Tang, X., Choi, Y., Heng, P.-A., Wu, F., & Zhang, Y. (2025). *RiboPO: Preference optimization for structure- and stability-aware RNA design* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2510.21161>

- Taneda, A. (2011). MODENA: a multi-objective RNA inverse folding. *Advances and Applications in Bioinformatics and Chemistry*, 4, 1–12. <https://doi.org/10.2147/AABC.S14335>
- Tinoco, I., Jr., & Bustamante, C. (1999). How RNA folds. *Journal of Molecular Biology*, 293(2), 271–281. <https://doi.org/10.1006/jmbi.1999.3001>
- Vandivier, L. E., Anderson, S. J., Foley, S. W., & Gregory, B. D. (2016). The conservation and function of RNA secondary structure in plants. *Annual Review of Plant Biology*, 67, 463–488. <https://doi.org/10.1146/annurev-arplant-043015-111754>
- Wachsmuth, M., Findeiß, S., Weissheimer, N., Stadler, P. F., & Mörl, M. (2013). De novo design of a synthetic riboswitch that regulates transcription termination. *Nucleic Acids Research*, 41(4), 2541–2551. <https://doi.org/10.1093/nar/gks1330>
- Yang, X., Yoshizoe, K., Taneda, A., & Tsuda, K. (2017). RNA inverse folding using Monte Carlo tree search. *BMC Bioinformatics*, 18(1), 468. <https://doi.org/10.1186/s12859-017-1882-7>
- Zhang, H., Zhang, L., Lin, A., Xu, C., Li, Z., Liu, K., et al. (2023). Algorithm for optimized mRNA design improves stability and immunogenicity. *Nature*, 621(7978), 396–403. <https://doi.org/10.1038/s41586-023-06127-z>
- Zuker, M., & Stiegler, P. (1981). Optimal computer folding of large RNA sequences using thermodynamics and auxiliary information. *Nucleic Acids Research*, 9(1), 133–148. <https://doi.org/10.1093/nar/9.1.133>